Gaussian Binomial Coefficients and the q-Binomial Theorem

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1 Motivation

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ are known as binomial coefficients. This is the vintage form of the binomial theorem that has been known since time immemorial. In this note, I would like to give an introduction to a generalised form of this theorem, known as the q-binomial theorem, whose coefficients are known as the Gaussian binomial coefficients or the q-binomial coefficients.

The motivation for this will be through a combinatorial problem. Consider the 2-dimensional $X-Y$ plane in Cartesian coordinates with all its lattice points. Draw all the lines passing through the lattice points that are parallel to either of the coordinate axes. We'll refer to these as lattice lines. Suppose h and k are 2 non-negative integers. Then, what are the number of ways of going from $(0,0)$ to (h,k) such that we move only along lattice lines and our each move occurs in either the positive X-direction (which we'll denote by R) or the positive Y-direction (which we'll denote by U)?

So, this is a quite standard problem in elementary counting theory and the answer is $\binom{h+k}{k}$. We'll try to visualise this solution with the help of the binomial theorem. Consider:

$$(R+U)^{h+k} = (R+U)(R+U)\ \ldots\ (R+U)$$

Any move sequence consisting of $h+k$ moves that consists of exactly h R's and k U's, which denote unit movements in the right and the up directions respectively, will take us from $(0,0)$ to (h,k). Any such sequence can be visualised as being obtained by choosing either R's or U's from each term of the RHS of the above equation such that we ensure that we pick the required number of each. Hence, the number of such sequences will equal the coefficient of $R^h U^k$ in the above expansion, which is equal to $\binom{h+k}{k}$ by the binomial theorem.

Now, let us look at a variant of this problem. Suppose we still want to go from $(0,0)$ to (h,k) using the same rules, but with the added rule that the area enclosed between our path and the X-axis should be exactly j units where j is an integer satisfying $0 \leq j \leq hk$. How many such paths are there now? If we take $j=0$, then there's only one such path - go from $(0,0)$ to $(h,0)$ and then to (h,k). How do we solve the problem for a general j?

2 The theorem

In order to move towards generalising the binomial theorem, we observe the behavior of x and y. Consider $(x+y)^2 = x(x+y)+y(x+y) = x^2+xy+yx+y^2 = x^2+2xy+y^2$. An underlying assumption used in this computation is that $xy = yx$ i.e., multiplication is commutative. What if we do away with this property of multiplication? What if we make $yx = q.xy$ for some q? We get the q-binomial theorem!

As non-commutative multiplication is a bit counter-intutive, it will be helpful to keep the following interpretation of x, y and q in mind throughout. Let q be a complex number such that $q \neq 0, 1$. Consider $x = \begin{pmatrix} 1 & 0 \\ 0 & q \end{pmatrix}$ and $y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ Then, it can be seen that $yx = q.xy$. I mention that there is

nothing natural about these definitions and that their purpose is just to provide an easy visualisation of our computations, and there can always be alternate definitions. With these definitions, and the observations that $qx = xq$ and $qy = yq$, we are now ready to establish the q-binomial theorem. So, we wish to establish an expression of the form:

$$(x+y)^n = \sum_{k=0}^{n} c(n,k)x^{n-k}y^k$$

where we need to determine $c(n,k)$. We'll try to solve this problem inductively. We try to expand $(x+y)^{n+1}$ in 2 ways. Firstly:

$$\begin{aligned}
&(x+y)^{n+1} = (x+y)^n(x+y)\\
&= \sum_{k=0}^{n} c(n,k)x^{n-k}y^k x + \sum_{k=0}^{n} c(n,k)x^{n-k}y^{k+1}\\
&= \sum_{k=0}^{n} c(n,k)q^k x^{n+1-k}y^k + \sum_{k=0}^{n} c(n,k)x^{n-k}y^{k+1}
\end{aligned}$$

Next, we also have:

$$\begin{aligned}
&(x+y)^{n+1} = (x+y)(x+y)^n\\
&= \sum_{k=0}^{n} c(n,k)x^{n-k+1}y^k + \sum_{k=0}^{n} c(n,k)yx^{n-k}y^k\\
&= \sum_{k=0}^{n} c(n,k)x^{n+1-k}y^k + \sum_{k=0}^{n} c(n,k)q^{n-k}x^{n-k}y^{k+1}
\end{aligned}$$

Comparing the coefficients of $x^{n+1-k}y^k$ in the 2 expressions obtained, we have upon simplification:

$$c(n,k) = \frac{1-q^{n+1-k}}{1-q^k}c(n,k-1)$$

Next, we define the following analogues for integers, factorials and binomial coefficients:

$[n]_q = 1 + q + q^2 + \ldots + q^{n-1}$
$n!_q = [n]_q[n-1]_q \ldots [2]_q[1]_q$
$\binom{n}{k}_q = \frac{n!_q}{k!_q(n-k)!_q}$

With these definitions in hand, we have $c(n,k) = \frac{[n+1-k]_q}{[k]_q}c(n,k-1)$. Thus, solving recursively and using $c(n,0) = 1$, we have:

$$c(n,k) = \binom{n}{k}_q = \frac{n!_q}{k!_q(n-k)!_q}$$

where $n!_q = 1(1+q)(1+q+q^2)\ldots(1+q+q^2\ldots+q^{n-1})$. Thus, we have the q-binomial theorem in its full glory as all of its coefficients have now been determined in a closed form.

It is interesting to notice the similarity between the usual binomial coefficients and their q-analogues. Also, it is easy to see that if we take the limits of all these expressions as q goes to 1, all the above definitions get reduced to their usual forms and we have our original binomial theorem as a special case of the q-binomial theorem.

3 Interpretations

It is quite amazing how the Gaussian Binomial Coefficients crop up in various fields in a completely unexpected manner. I'll be giving a few examples here to demonstrate this fact.

First we look at linear algebra. Suppose we have an n-dimensional vector space V over a finite field

$\mathbb{F}_q$. For a given $k \leq n$, how many k-dimensional subspaces does V have? We'll tackle this problem using the following observation: The number of k-dimensional subspaces of V multiplied by the number of ordered bases of each subspace will give us the number of ordered bases of V consisting of k elements.

Now, let W be a k-dimensional subspace. Then, it consists of q^k elements. The first element of a basis can be selected in $q^k - 1$ ways (as we don't want 0 in our basis). The next element shouldn't be a scalar multiple of the first one, and so can be selected in $q^k - q$ ways. The next one shouldn't be linear combination of the first two, and so, can be selected in $q^k - q^2$ ways. Going on, we get that for a given subspace of dimension k, the number of ordered bases is $(q^k - 1)(q^k - q) \ldots (q^k - q^{k-1})$.

Proceeding in exactly the same manner, the number of ordered bases of V of size k will be equal to $(q^n - 1)(q^n - q) \ldots (q^n - q^{k-1})$. Thus, making use of our observation above, we have that the number of k-dimensional subspaces of V is given by:

$$\frac{(q^n - 1)(q^n - q)...(q^n - q^{k-1})}{(q^k - 1)(q^k - q)...(q^k - q^{k-1})} = \frac{(q^n - 1)(q^{n-1} - 1)...(q^{n-k+1} - 1)}{(q^k - 1)(q^{k-1} - 1)...(q - 1)} = \binom{n}{k}_q$$

and thus, we have a natural occurrence of q-binomial coefficients here.

Next, we look at a tiling interpretation for these coefficients. Consider a board of length n that needs to be tiled using k turquoise squares and $n - k$ magenta squares. Then, the number of ways of doing this is exactly $\binom{n}{k}$. Now, suppose we introduce weights on a tiling as follows. Each square of a tiling is assigned a weight. Each magenta square has weight 1. Each turquoise square has weight q^s where s denotes the number of magenta squares appearing to the left of the given turquoise square. The net weight of the tiling is calculated by multiplying the weights of all the individual squares.

Then, if we denote by $\mathcal{T}_{n,k}$ the set of all tilings of an n-board with k turquoise squares and $n - k$ magenta squares and if $w(T)$ denotes the weight of a tiling T, then we have the identity:

$$\binom{n}{k}_q = \sum_{T \in \mathcal{T}_{n,k}} w(T)$$

The proof of this fact will become clear in section 6 and so, we assume this identity to be true for now. Hence, the above 2 examples show that the q-binomial coefficients are omnipresent in Combinatorics.

4 Identities

Using the tiling interpretation of the previous section, we can provide proofs of q-analogues for several combinatorial identities involving the ordinary binomial coefficients. I provide a few examples here. It is interesting to see that the proving style exactly matches the method used for proving these identities for the ordinary binomial coefficients.

First of all, we try to prove $\binom{n}{k}_q = \binom{n}{n-k}_q$. Now, the LHS denotes the total weight of all the tilings of an n-board using k turquoise squares. Consider any such tiling. If we toggle the colours of this tiling and reverse the order of the tiling, it is easy to see that we get a tiling of an n-board using $n - k$ turquoise squares with the same weight as the original tiling. Hence, we have a bijection between the 2 sets and so, their net weights must match. Thus, we have the required identity.

Next, we prove $\binom{n}{k}_q = \binom{n-1}{k}_q + \binom{n-1}{k-1}_q q^{n-k}$. The LHS denotes the total weight for the tilings of an n-board using k turquoise squares. Now, in any tiling, the last square is either turquoise or magenta. If it is magenta, removing that last tile (which provides weight 1), we are left with a tiling of an $(n-1)$-board using k turquoise squares whose net weight is $\binom{n-1}{k}_q$. If the last square is turquoise, on removing it, we are left with a tiling of an $(n-1)$-board using $k-1$ turquoise squares whose net weight is $\binom{n-1}{k-1}_q$. This weight, when multiplied by q^{n-k} (the weight of the last turquoise square) and

added to the previous weight, should denote the net weight of such tilings which is $\binom{n}{k}_q$ and hence, the identity stands proven.

In exactly the same manner as above, if we consider the cases for the first square of the tiling instead of the last square, we get the identity $\binom{n}{k}_q = \binom{n-1}{k-1}_q + \binom{n-1}{k}_q q^k$.

Finally, we provide a proof for the identity: $\binom{2n}{n}_q = \sum_{j=0}^{n} q^{j^2} \binom{n}{j}_q^2$. The LHS denotes the net weight of all the tilings of a $2n$-board using n turquoise squares. Now, let's calculate this weight in a different manner. Suppose the first n squares consist of $n-j$ turquoise squares and the final n consist of j turquoise squares. Then, if we consider them in an independent manner, the net contribution to weight from the first n square is $\binom{n}{n-j}_q = \binom{n}{j}_q$ and the last n is $\binom{n}{j}_q$. But the weight contribution of each of the j turquoise squares among the last n squares will also be affected by the j magenta squares occurring among the first n squares, thus making a total of $(q^j)^j = q^{j^2}$. Thus, the total weight becomes $q^{j^2}\binom{n}{j}_q^2$ and when we sum over j, we get what we set out to prove.

5 The path counting problem

Now, let us return to the problem that motivated our journey. To recall, we need to find the number of ways of going from $(0,0)$ to (h,k), $h,k \geq 0$, moving along lattice lines only, either rightwards or upwards, in a manner such that the area enclosed between the path and the X-axis is exactly j units where $0 \leq j \leq hk$. Now, let us look at:

$$(R+U)^{h+k} = (R+U)(R+U)\ \ldots\ (R+U)$$

In the earlier case of the problem, it didn't matter if we went up first and then right or vice-versa, the order didn't affect our calculations. In algebraic terms, earlier we had $RU = UR$ and so we had been able to use our traditional binomial theorem. In this problem, RU and UR will be different because these paths enclose different areas beneath them. If we put $R = x$ and $U = y$ where the notation is as used in section 2, we'll have $UR = q.RU$. In fact, given any string consisting of U's and R's, it can be reduced to a canonical form $q^t.R^uU^v$ where $t,u,v \in \mathbb{Z}^+ \cup \{0\}$. For example, $URURUU = URq.RUUU = Uq.RRUUU = q.URRUUU = q.q.RURUUU = q^2.RURUUU = q^2.Rq.RUUUU = q^2.q.RRUUUU = q^3.R^2U^4$.

The above string corresponds to a move from $(0,0)$ to $(2,4)$ where the area enclosed between the path and the X-axis is 3 units. Looking at the reduction process geometrically, everytime we replace UR by RU, the area enclosed gets reduced by 1 and at the same time, a q gets introduced in the term. Therefore, it is easy to see that every string of U's and R's can be reduced to a unique such canonical form where the index of q denotes the area enclosed beneath the path. (A rigorous justification for this statement can be provided using Diamond Lemma, but we'll assume the result implicitly for the sake of simplicity. Interested readers may refer to [3])

Hence, we can can conclude that if we expand $(R+U)^n$ using the q-binomial theorem rules, then the coefficient of $q^aR^bU^c$, where $b+c=n$, denotes the number of paths from $(0,0)$ to (b,c) that enclose an area of a units below the path. Thus, we have a solution to our problem.

In order to demonstrate the working of the solution, I state an example. Suppose we wish to go from $(0,0)$ to $(4,5)$ with an area of 12 units enclosed by the path. By the q-binomial theorem, we have:

$$(R+U)^9 = \sum_{k=0}^{9} \binom{9}{k}_q R^{n-k}U^k$$

Coefficient of R^4U^5 is given by $\binom{9}{4}_q = (1-q^9)(1-q^8)(1-q^7)(1-q^6)/(1-q^4)(1-q^3)(1-q^2)(1-q) = q^{20}+q^{19}+2q^{18}+3q^{17}+5q^{16}+6q^{15}+8q^{14}+9q^{13}+11q^{12}+11q^{11}+12q^{10}+11q^9+11q^8+9q^7+8q^6+6q^5+5q^4+3q^3+2q^2+q+1$. Thus, the required answer is the coefficient of $q^{12}R^4U^5 = 11$. Therefore,

the number of paths is 11.

Another elegant interpretation of these coefficients is as follows: What are the number of partitions of an integer n, when there are at most h parts and each part is at most k (given that $n \leq hk$)? This problem is equivalent to finding the number of paths from $(0,0)$ to (h,k) where the area enclosed below the path is n units. This is because given any such path, we can visualise the area beneath it as a union of columns where each column can be looked upon as a part, the sum of all the parts being n. Conversely, given a partition, if we arrange parts in an increasing order and construct a column corresponding to each part, we get a path to (h,k) from $(0,0)$ enclosing a total area of n units. Hence, there is a bijection between the 2 sets and so they have the same cardinalities.

6 The infinite q-binomial theorem

Apart from the form given here, the q-Binomial Theorem is often given in another form:

$$(1+x)(1+qx)...(1+q^{n-1}x) = \sum_{k=0}^{n} \binom{n}{k}_q q^{k(k-1)/2} x^k$$

where x and q are complex numbers. The main advantage of using this form of the theorem is that here, we don't have to bother about non-commutative multiplication and so, we can work in the familiar realm of $\mathbb{C}$. The proof of this form can be obtained exactly in the same manner as we proved the other form of the theorem, by expanding inductively from both directions of the LHS.

Another advantage of this form is that it leads to an easy generalisation to an infinite q-Binomial Theorem. We proceed as follows. First of all, we take the limit of the above expression as n goes to ∞. Then if we use the notation $(q)_k = (1-q)(1-q^2)...(1-q^k)$, we get

$$\prod_{k=0}^{\infty}(1+q^k x) = \sum_{k=0}^{\infty} \frac{q^{k(k-1)/2}x^k}{(q)_k}$$

In the above, the exchange of limit and summation on the RHS is justified by the uniform convergence of the power series for a suitable radius of convergence, given $|q| < 1$. Now, if we put $x = -1$ in the finite form of the theorem, we get:

$$\sum_{k=0}^{n} \frac{q^{k(k-1)/2}(-1)^{n-k}}{(q)_k (q)_{n-k}} = 0$$

for all $n > 0$ and for $n = 0$, LHS=1. Therefore, if we define $A(x) = \sum_{k=0}^{\infty} \frac{q^{k(k-1)/2}}{(q)_k} x^k$ and $B(x) = \sum_{k=0}^{\infty} \frac{(-x)^k}{(q)_k}$, then the product of these 2 power series must equal 1. Hence, we get:

$$\sum_{k=0}^{\infty} \frac{q^{k(k-1)/2}}{(q)_k} x^k \sum_{k=0}^{\infty} \frac{(-x)^k}{(q)_k} = 1$$

or

$$\prod_{k=0}^{\infty} \frac{1}{(1+q^k x)} = \sum_{k=0}^{\infty} \frac{(-x)^k}{(q)_k}$$

These 2 identities that express an infinite product as an infinite sum are known as Euler's identities. Now, put $x = -a$ in the finite form of the theorem to get:

$$\frac{(1-a)(1-qa)...(1-q^{n-1}a)}{(q)_n} = \sum_{k=0}^{n} \frac{q^{k(k-1)/2}(-a)^k}{(q)_k (q)_{n-k}}$$

Putting the LHS to be a_n, if we define a power series of the form $\sum_{k=0}^{\infty} a_k x^k$, then using the above identity, we have:

$$\sum_{k=0}^{\infty} \frac{(1-a)(1-qa)...(1-q^{k-1}a)}{(q)_k} x^k = \sum_{k=0}^{\infty} \frac{q^{k(k-1)/2}}{(q)_k} (-ax)^k \sum_{k=0}^{\infty} \frac{x^k}{(q)_k}$$

And thus, by Euler's identities, we have our final form of the infinite q-binomial theorem:

$$\sum_{k=0}^{\infty} \frac{(1-a)(1-qa)...(1-q^{k-1}a)}{(q)_k} x^k = \prod_{k=0}^{\infty} \frac{(1-q^k ax)}{(1-q^k x)}$$

Unsurprisingly, if we replace a by q^a and try to find the limit of the above expression as q tends to 1^-, where a is an integer, we have:

$$\sum_{k=0}^{\infty} \frac{(a)(a+1)...(a+k-1)}{k!} x^k = (1-x)^{-a}$$

which is exactly the traditional binomial series.

References

[1] The q-Binomial Theorem - Shaun Cooper, 2013

[2] A Tiling Interpretation of q-Binomial Coefficients - Jonathan Azose, 2007

[3] The Diamond Lemma for Ring Theory - George M. Bergman, 1978

[4] The q-Binomial Theorem - Heng Huat Chan, Song Heng Chan, Shaun Cooper, 2006